Oleh Boloban

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I have worked across medical, fintech, and AI-driven domains, where I was responsible for designing scalable, high-load systems. My task was to build reliable microservice architectures, integrate advanced AI solutions, and establish end-to-end ML/MLOps pipelines. I achieved this by developing distributed systems with Node.js and RabbitMQ, implementing CI/CD automation, and integrating Claude, AWS Bedrock, and SageMaker into production workflows. I also applied research in biomedical signal processing and telemedicine to create data-driven healthcare solutions. As a result, I delivered robust, efficient platforms that improved system performance, automated workflows, and enabled smarter decision-making across products.

EXPERIENCE

RiseApps *Bucharest, Romania*
Software engineer *01 July 2025 - Present*

- Designed and developed a microservice-based medical system using Node.js, implementing secure REST APIs, role-based access control, and integration with SQL/NoSQL databases.
- Implemented asynchronous communication between services using RabbitMQ, including queues, routing, retries, message deduplication, and event-driven workflows.
- Built and maintained CI/CD pipelines (Docker, GitHub/GitLab CI, Kubernetes/ECS) to automate testing, deployment, and scaling of microservices.
- Integrated AI systems using Claude API, enabling intelligent assistants, document processing workflows, and advanced conversational capabilities across internal tools.
- Created and managed a company-wide Knowledge Base, implementing RAG (Retrieval-Augmented Generation) with semantic search and automated document indexing.
- Developed and optimized prompt engineering strategies, improving LLM reliability, context handling, response accuracy, and system/user prompt structures.
- Implemented MLOps practices, including experiment tracking (MLflow/W&B), data preprocessing pipelines, model versioning, and automated model training/testing workflows.
- Built production-grade ML pipelines, with monitoring for data drift, model degradation, automated alerts, and scalable inference infrastructure.
- Developed AI solutions using AWS Bedrock, integrating managed LLMs into products and building high-availability inference endpoints.
- Used AWS SageMaker for ML training and deployment, configuring training jobs, SageMaker Pipelines, scalable inference endpoints, cost optimization, and integration with S3, Lambda, API Gateway, and CloudWatch.

Technical University of Ukraine 'Kyiv Polytechnic Institute' *Kyiv, Ukraine*
Research Assistant *01 September 2021 - Present*

- Microservices and service-oriented architecture: Designed and implemented scalable systems leveraging event-driven and edge-computing architectures to process real-time data.
- AI model development: Designed and trained machine learning models for analyzing patient health metrics, enabling data-driven insights and early anomaly detection.
- Edge computing & event-driven architecture: Implemented distributed systems with localized processing at the edge, enabling faster, low-latency decision-making for critical health monitoring.
- Research in telemedicine: Conducted studies on remote patient monitoring and data transmission, focusing on optimizing system performance and reliability.
- Course development and teaching: Designed and delivered a web development course for undergraduate students, covering both theory and practical programming skills.
- This role provided a multidisciplinary experience combining embedded systems, AI, and healthcare, contributing to the development of innovative telemedicine solutions while mentoring the next generation of developers.

- Publications: https://orcid.org/0009-0004-9074-4077

SimplifyLabs *Kingstown, St. Vincent and the Grenadines*
Software engineer *27 May 2024 – 30 June 2025*

- Worked in high-load environments utilizing modern technologies and a microservices-based architecture.
- Development of a Telegram bot: Designed and maintained the bot using Nest.js, integrating it with external services and internal modules.
- Developed a system for virtual payment cards linked to cryptocurrency wallets, integrating payment gateways and crypto services.
- Designed architectural solutions for crypto projects using microservices and event-driven architecture, enhancing flexibility and scalability.

- Managed the full product lifecycle, including planning, implementation, service expansion, and process management within the cryptocurrency system, along with integrating new functional modules.
- This experience allowed me to tackle complex technical challenges while working in a dynamic blockchain environment, optimizing processes and expanding product functionality.

Brightgrove *Kyiv, Ukraine*
Javascript Engineer *07 November 2021 – 02 May 2024*

- Worked on building data-driven solutions for video content scheduling across multiple platforms, leveraging serverless computing and cloud-based architectures. Key responsibilities and achievements include:
- AWS Lambda development: Built and maintained AWS Lambda functions to generate video content schedule files for various platforms, optimizing for scalability and performance.
- Data processing optimization: Adapted data processing algorithms to fit within AWS Lambda's execution and memory constraints, ensuring efficient processing of large datasets.
- Complex SQL queries: Designed and optimized advanced SQL queries to retrieve and manipulate data in PostgreSQL, meeting complex business logic requirements.
- Data migration system: Developed a robust data migration pipeline to transfer and
- transform data from MongoDB to PostgreSQL, ensuring consistency and reliability during the process.
- Template-based file generation: Implemented a system of dynamic templates for different platforms, generating data-rich files based on template specifications and database inputs.
- Cloud tools and services: Utilized AWS CodeBuild and S3 for deployment automation, artifact storage, and seamless integration with serverless environments.

SoftServe *Kyiv, Ukraine*
SoftwareEngineer *12 December 2020 – 21 October 2021*

- Developing financial logic for high-load transactional processes within fintech applications, ensuring high accuracy and reliability of financial calculations.
- Integrating AWS Lambda with Salesforce to enable real-time data exchange and automate critical business processes.
- Designing and optimizing complex SQL queries to efficiently manage large datasets and support high-performance database operations.
- Building complex fintech calculations for recommendation engines and discount systems, leveraging dynamic financial models and scenario-based computations.
- Collaborating with cross-functional teams to ensure product scalability and alignment with business requirements.

Geniusee *Kyiv, Ukraine*
SoftwareEngineer *13 August 2018 – 02 December 2020*

- Developed and maintained web applications using both CMS platforms and native JavaScript tools.
- Designed and implemented scalable, maintainable web application architectures.
- Led a team of two developers, overseeing task delegation, code review, and delivery timelines.
- Applied native JavaScript techniques to optimizing performance and reduce dependency overhead.

EDUCATION

National Technical University of Ukraine "Igor Sikorsky Kyiv Polytechnic Institute" *Kyiv, Ukraine*
PhD of Computer Science *Sept 30, 2025*

- Research in eHealth, medical artificial intelligence for classification of apnea diseases and microservices architecture
- The member of government project #18.2021.MM

National Technical University of Ukraine "Igor Sikorsky Kyiv Polytechnic Institute" *Kyiv, Ukraine*
Master of Computer Science *May 31, 2021*

- CGPA – 84 (B)
- Research in eHealth and methods of registration breath diseases

National Technical University of Ukraine "Igor Sikorsky Kyiv Polytechnic Institute" *Kyiv, Ukraine*
Bachelor of Science in Mechanical Engineering *Jun 30, 2019*

- CGPA – 73 (D)
- The member of Students Council

SKILLS

- **Technical stack:** JavaScript, TypeScript, SQL and NoSQL DB, Message brokers (Kafka, RabbitMQ, NATS), AWS, Golang, Python, Docker, Kubernetes.
- **Soft skills:** communicative, proactive, work effectively in ambiguous environments, adapt quickly to change, and handle crisis situations with confidence and clarity.
- **Languages:** English (Upper intermediate), Russian (Speaking and Writing), Ukrainian (Fluent)